

Questions

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Q1 - 2024 (04 Apr Shift 1)

Let $f(x) = x^5 + 2e^{x/4}$ for all $x \in \mathbf{R}$. Consider a function $g(x)$ such that $(g \circ f)(x) = x$ for all $x \in \mathbf{R}$. Then the value of $8g'(2)$ is :

- (1) 2
- (2) 8
- (3) 4
- (4) 16

Q2 - 2024 (05 Apr Shift 2)

If $y(\theta) = \frac{2 \cos \theta + \cos 2\theta}{\cos 3\theta + 4 \cos 2\theta + 5 \cos \theta + 2}$, then at $\theta = \frac{\pi}{2}$, $y'' + y' + y$ is equal to :

- (1) $\frac{1}{2}$
- (2) 1
- (3) 2
- (4) $\frac{3}{2}$

Q3 - 2024 (06 Apr Shift 1)

If $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ then

- (1) $f''\left(\frac{2}{\pi}\right) = \frac{24-\pi^2}{2\pi}$
- (2) $f''\left(\frac{2}{\pi}\right) = \frac{12-\pi^2}{2\pi}$
- (3) $f''(0) = 1$
- (4) $f''(0) = 0$

Q4 - 2024 (06 Apr Shift 2)

Suppose for a differentiable function h , $h(0) = 0$, $h(1) = 1$ and $h'(0) = h'(1) = 2$. If $g(x) = h(e^x)e^{h(x)}$, then $g'(0)$ is equal to:

- (1) 5

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(2) 4

(3) 8

(4) 3

Q5 - 2024 (09 Apr Shift 1)

Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to:

(1) 73

(2) 62

(3) 51

(4) 54

Q6 - 2024 (09 Apr Shift 2)

If $\log_e y = 3 \sin^{-1} x$, then $(1 - x^2) y'' - xy'$ at $x = \frac{1}{2}$ is equal to

(1) $3e^{\pi/6}$

(2) $9e^{\pi/2}$

(3) $3e^{\pi/2}$

(4) $9e^{\pi/6}$

Questions

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Answer Key

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Q1 (4) mathongo Q2 (3) mathongo Q3 (1) mathongo Q4 (2) mathongo

Q5 (3) mathongo Q6 (2) mathongo mathongo mathongo mathongo mathongo

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Solutions

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Q1

$$f(x) = 2$$

when $x = 0$

$$\therefore g'(f(x))f'(x) = 1$$

$$g'(2) = \frac{1}{f'(0)}$$

$$\therefore f'(x) = 5x^4 + \frac{2}{4}e^{x/4}$$

$$g'(2) = 2$$

Ans = 16

Option (1)

Q2

$$y = \frac{2 \cos \theta + 2 \cos^2 \theta - 1}{4 \cos^3 \theta - 3 \cos \theta + 8 \cos^2 \theta - 4 + 5 \cos \theta + 2}$$

$$(2 \cos^2 \theta + 2 \cos \theta - 1)$$

$$y = \frac{1}{(2 \cos^2 \theta + 2 \cos \theta - 1)(2 \cos \theta + 2)}$$

$$y = \frac{1}{2} \left(\frac{1}{1 + \cos \theta} \right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$y' = \frac{1}{2} \left(\frac{-1}{(1 + \cos \theta)^2} \times (-\sin \theta) \right)$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = \frac{1}{2}$$

$$y'' = \frac{1}{2} \left[\frac{\cos \theta (1 + \cos \theta)^2 - \sin \theta (2)(1 + \cos \theta)(-\sin \theta)}{(1 + \cos \theta)^4} \right]$$

$$\Rightarrow \theta = \frac{\pi}{2} \quad y = 1$$

Q3

$$f'(x) = 3x^2 \sin\left(\frac{1}{x}\right) - x \cos\left(\frac{1}{x}\right)$$

$$f''(x) = 6x \sin\left(\frac{1}{x}\right) - 3 \cos\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) - \frac{\sin\left(\frac{1}{x}\right)}{x}$$

$$f''\left(\frac{2}{\pi}\right) = \frac{12}{\pi} - \frac{\pi}{2} = \frac{24 - \pi^2}{2\pi}$$

Q4

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Solutions

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$$g(x) = h(e^x) \cdot e^{h(x)}$$

$$g'(x) = h(e^x) \cdot e^{h(x)} \cdot h'(x) + e^{h(x)} h'(e^x) \cdot e^x$$

$$g'(0) = h(1)e^{h(0)}h'(0) + e^{h(0)}h'(1) \\ = 2 + 2 = 4$$

Q5

$$f(x) = ax^3 + bx^2 + cx + 41$$

$$f'(x) = 3ax^2 + 2bx + cx$$

$$\Rightarrow f'(1) = 3a + 2b + c = 2 \dots (1)$$

$$f''(x) = 6ax + 2b$$

$$\Rightarrow f''(1) = 6a + 2b = 4$$

$$3a + b = 2 \dots \dots \dots (2)$$

$$(1) - (2)$$

$$b + c = 0 \dots (3)$$

$$f(1) = 40$$

$$a + b + c + 41 = 40$$

$$\text{use (3)}$$

$$a + 41 = 40$$

$$\text{by (2)}$$

$$-3 + b = 2 \Rightarrow b = 5 \& c = -5$$

$$a^2 + b^2 + c^2 = 1 + 25 + 25 = 51$$

Q6

$$\ln(y) = 3 \sin^{-1} x$$

$$\frac{1}{y} \cdot y' = 3 \left(\frac{1}{\sqrt{1-x^2}} \right)$$

$$\Rightarrow y' = \frac{3y}{\sqrt{1-x^2}} \text{ at } x = \frac{1}{2}$$

$$\Rightarrow y' = \frac{3e^{3\left(\frac{\pi}{6}\right)}}{\frac{\sqrt{3}}{2}} = 2\sqrt{3}e^{\frac{\pi}{2}}$$

$$\Rightarrow y'' = 3 \left(\frac{\sqrt{1-x^2}y' - y \frac{1}{2\sqrt{1-x^2}}(-2x)}{(1-x^2)} \right)$$

$$\Rightarrow (1-x^2)y'' = 3 \left(3y + \frac{xy}{\sqrt{1-x^2}} \right)$$

$$\downarrow \text{ at } x = \frac{1}{2}, y = e^{3 \sin^{-1}\left(\frac{1}{2}\right)} = e^{3\left(\frac{\pi}{6}\right)} = e^{\frac{\pi}{2}}$$

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Solutions

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$$\begin{aligned}
 (1-x^2)y'' \Big|_{\text{at } x=\frac{1}{2}} &= 3 \left(3e^{\frac{\pi}{2}} + \frac{\frac{1}{2} \left(e^{\frac{\pi}{2}} \right)}{\frac{\sqrt{3}}{2}} \right) \\
 &= 3e^{\frac{\pi}{2}} \left(3 + \frac{1}{\sqrt{3}} \right) \\
 (1-x^2)y'' - xy' \Big|_{\text{at } x=\frac{1}{2}} \\
 &= 3e^{\frac{\pi}{2}} \left(3 + \frac{1}{\sqrt{3}} \right) - \frac{1}{2} \left(2\sqrt{3}e^{\frac{\pi}{2}} \right) = 9e^{\frac{\pi}{2}}
 \end{aligned}$$

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